DAY 1 – Cybersecurity, SOC & Git Basics

Objective

By the end of this session, students will be able to:

- Understand the fundamental concepts of cybersecurity
- Explain the role and importance of a Security Operations Center (SOC)
- Identify and analyze common cyber attacks
- Use basic Git commands to manage and upload projects

PART 1: What is Cybersecurity?

Cybersecurity is the practice of protecting:

- Systems (computers, servers)
- Networks (internet, Wi-Fi)
- Data (personal, financial, organizational)

from:

- Unauthorized access
- Cyber attacks
- Data breaches

Core Concept (CIA Triad)

Cybersecurity is built on three key principles:

- **Confidentiality** → Data should be accessed only by authorized users
- **Integrity** → Data should not be altered or tampered with
- **Availability** → Systems should be accessible when needed

Real-Life Examples

1. Social Media Hacking

- Weak passwords or reused credentials
- Attacker gains full account access

2. Bank Fraud (OTP Scam)

- User receives fake call or SMS
- Shares OTP unknowingly
- Money gets transferred

3. Phishing Emails

- Fake emails pretending to be trusted sources
- Trick users into entering credentials

4. Malware Attacks

- User installs infected software
- System gets compromised

Real-World Scenario

A user reports:

"My bank account was accessed without permission."

Security Perspective:

- Was phishing involved?
- Was malware installed?
- Were credentials leaked?

PART 2: Types of Cyber Attacks

1. Phishing Attack

Phishing is a **social engineering attack** where attackers trick users into revealing sensitive information.

Example:

Email:

“Your account will be suspended. Click here to verify.”

Indicators of Phishing:

- Suspicious sender email
- Urgent or threatening language
- Fake or misspelled domain
- Unknown links

Prevention:

- Always verify URLs
- Avoid clicking unknown links
- Enable Multi-Factor Authentication (MFA)

2. Malware Attack

Malware is malicious software designed to harm systems or steal data.

Types of Malware

Type	Description
Virus	Attaches to files
Worm	Spreads automatically
Trojan	Disguised as legitimate software
Ransomware	Locks data and demands payment

Real Scenario:

- User downloads cracked software
- Malware installs silently
- Attacker gains remote access

3. Brute Force Attack

An attack where multiple password combinations are tried until the correct one is found.

Example:

- Trying passwords like:
  - 123456
  - password
  - admin123

Prevention:

- Strong passwords
- Account lockout mechanisms
- CAPTCHA
- MFA

PART 3: What is SOC?

A Security Operations Center (SOC) is a centralized team responsible for:

- Monitoring systems
- Detecting cyber threats
- Responding to incidents

Simple Analogy:

SOC Component	Real-Life Equivalent
Logs	CCTV footage
Analyst	Security guard
Alerts	Alarm system

SOC Responsibilities

1. Monitoring

    - Continuous monitoring of systems and logs

2. Detection

    - Identifying suspicious activities

3. Response

    - Taking action against threats

4. Analysis

    - Understanding attack patterns
    - Improving security

SOC Workflow

1. Logs are generated
2. SIEM tools collect logs
3. Alerts are triggered
4. Analysts investigate
5. Response actions are taken

Real SOC Scenario

Alert:

"Multiple failed login attempts detected"

Analyst Investigation:

- Check IP address
- Verify user activity
- Analyze login patterns

Action:

- Block IP
- Reset password
- Report incident

PART 4: Activity – Phishing Detection

Phishing is a type of cyber attack where an attacker pretends to be a trusted person or organization to trick you into giving sensitive information like:

- Passwords
- OTPs
- Bank details

What to Check in a Link or Email ?

Basic Checks

- Sender email address
- Spelling and grammar

Advanced Checks

- Hover over links (check real URL)
- Domain name verification
- HTTPS presence
- Avoid unknown attachments

PART 5: Git Basics

What is Git?

Git is a version control system used to:

- Track changes in code
- Manage project history
- Collaborate with teams

Basic Git Commands

- Initialize Repository

git init

- Check Status

git status

- Add Files

git add .

- Commit Changes

git commit -m "first commit"

- Connect to Remote Repository

git remote add origin https://github.com/username/repo.git

- Push Code to GitHub

git push -u origin main

Git Workflow:

1. Create project folder
2. Initialize Git
3. Add files
4. Commit changes
5. Push to GitHub